

Data Cleaner Theory

Theory Guide: Data Preprocessing, Imputation & Outlier Handling

1. Theoretical Foundations of Missing Data

1.1 Missing Data Mechanisms

Missing data can be classified into three statistical mechanisms:

- Missing Completely at Random (MCAR):** Missingness is independent of both observed and unobserved data.
- Missing at Random (MAR):** Missingness depends on observed features but not on the missing value itself (e.g., older patients less likely to report certain metrics).
- Missing Not at Random (MNAR):** Missingness is directly related to the unobserved missing value itself.

2. Imputation Methodologies

Missing values can distort data distributions and impact machine learning models. Below are common imputation methods:

Imputation Method	Mathematical/Conceptual Basis	Key Advantages	Typical Use Cases
Median Imputation	Replaces missing values with the 50th percentile.	Robust to skewed distributions and outliers.	Continuous numerical variables with non-normal distributions (e.g., <code>bmi</code>).
Mode Imputation	Replaces missing values with the most frequent category.	Preserves original categorical labels.	Categorical variables (e.g., <code>gender</code> , <code>region</code>).
Missing Indicator	Adds a binary column ($I_{\text{missing}} \in \{0, 1\}$).	Retains missingness patterns for predictive models.	Variables where missingness itself is predictive (e.g., <code>age_missing</code>).
KNN Imputer	Uses weighted averages of k nearest neighbors.	Captures local feature relationships.	Multi-variable correlated continuous data (e.g., <code>cholesterol</code> , <code>glucose</code>).

MICE (Iterative Imputer)	Models each feature with missing values as a function of others iteratively.	Preserves inter-variable dependencies and variance structure.	Complex datasets with multivariate missingness.
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3. Outlier Detection & Treatment Techniques

Outliers are extreme data points that deviate significantly from the general pattern, potentially skewing results.

3.1 Detection Methods

1. **Z-Score Method (Parametric):**

- Formula: $Z = (x - \mu) / \sigma$
- Rule: Flag instances where $|Z| > 3$.
- Assumption: Data follows a Normal (Gaussian) distribution.

2. **Interquartile Range (IQR) Method (Non-Parametric):**

- Formula: $IQR = Q3 - Q1$
- Rule: Bounds are $[Q1 - 1.5 * IQR, Q3 + 1.5 * IQR]$.
- Assumption: Distribution-agnostic; suitable for skewed features.

3. **Percentile Capping Method:**

- Rule: Flags points below the 1st percentile or above the 99th percentile.
- Purpose: Restricts extreme anomalies without dropping rows.

3.2 Outlier Handling: Winsorization vs. Deletion

- **Row Removal:** Drops rows with extreme outliers. Risky for small datasets as it reduces sample size.
- **Winsorization (Capping):** Clamps extreme values at specified percentiles (e.g., 1st and 99th). Preserves dataset size while reducing variance.

4. Before-and-After Cleaning Metrics Summary

Data cleaning impacts key descriptive statistics:

- **Variance Reduction:** Capping extreme values lowers standard deviation σ while retaining central tendency measures (mean, median).
- **Completeness:** Imputation ensures 100% feature completeness (e.g., N=500 rows).
- **Feature Expansion:** Adding missingness indicators (e.g., `age_missing`) increases feature space dimensionality without data loss (e.g., N_features : 9 -> 10).

Data Cleaning Workflow

